

Chapter 26. Multivariable Calculus for Applied Mathematics

Fields, flows, conservation laws, and numerical models

Applied mathematics is the art of turning a real system into a mathematical model, studying the model, computing with it, and interpreting the result. Multivariable calculus is one of the main languages of this process. It tells us how quantities change in space and time, how material flows through regions, how heat diffuses, how forces produce motion, and how energy principles lead to equations.

The previous chapters developed gradients, Jacobians, Hessians, multiple integrals, vector fields, line integrals, surface integrals, and the major integral theorems. This chapter brings those ideas together from an applied mathematics point of view. The goal is not to replace a future course in differential equations, numerical analysis, or partial differential equations. The goal is to show how multivariable calculus becomes the foundation for those subjects.

In this chapter we study four recurring applied themes:

1. **Local change:** derivatives measure how quantities change at a point.
2. **Accumulation:** integrals measure total amount over curves, surfaces, and regions.
3. **Balance laws:** divergence and flux express conservation of mass, heat, probability, or charge.
4. **Energy principles:** many equations arise by minimizing an energy.

Learning goals

After completing this chapter, you should be able to:

- interpret scalar fields, vector fields, gradient fields, divergence, curl, and the Laplacian in applied models;
- derive simple conservation laws from flux balance;
- connect the heat equation to diffusion, smoothing, and gradient-based dynamics;
- use finite differences to approximate gradients, divergence, and the Laplacian on a grid;
- explain how continuous models lead to numerical algorithms;
- recognize variational and energy-based formulations;
- connect applied calculus ideas to data science, image processing, probability flow, and optimization.

1 The applied mathematics pipeline

A useful applied mathematics workflow is

system $\longrightarrow$ variables $\longrightarrow$ model $\longrightarrow$ analysis $\longrightarrow$ computation $\longrightarrow$ interpretation.

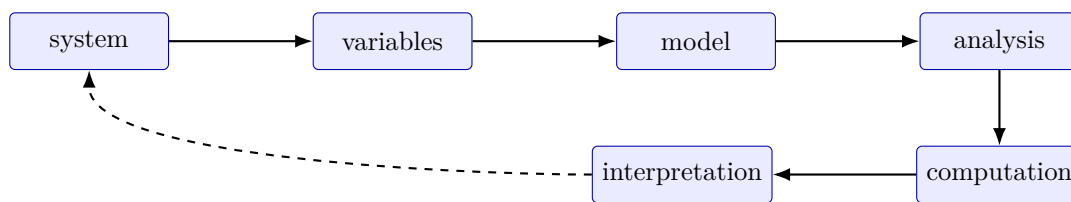


Figure 1: The applied mathematics pipeline. Interpretation often feeds back into the choice of variables and model.

For example, if $u(x, t)$ is the temperature of a thin rod at position x and time t , a physical law says heat flows from hot regions to cold regions. A calculus model converts this statement into a differential equation. A numerical method approximates the equation on a finite grid, and a simulation helps us understand the temperature evolution.

Applied setting	Typical field	Calculus object
heat flow	$u(x, y, t)$	scalar field
fluid velocity	$F(x, y, z, t)$	vector field
probability density	$p(x, t)$	scalar field with integral 1
image intensity	$I(x, y)$	scalar field on a pixel grid
model loss	$L(\theta)$	scalar field on parameter space
gradient descent	$\theta'(t) = -\nabla L(\theta)$	vector field on parameter space

A key message of this chapter is that the same calculus operators appear across many disciplines.

2 Scalar fields, vector fields, and operators

A scalar field assigns a number to each point. Examples include temperature, pressure, density, altitude, potential energy, image brightness, or a loss function:

$$u = u(x, y, z).$$

A vector field assigns a vector to each point. Examples include velocity, force, heat flux, electric field, gradient direction, or parameter update direction:

$$F(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle.$$

The most important differential operators in applied mathematics are

$$\nabla u = \langle u_x, u_y, u_z \rangle, \quad \nabla \cdot F = P_x + Q_y + R_z,$$

$$\nabla \times F = \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle, \quad \Delta u = \nabla \cdot (\nabla u) = u_{xx} + u_{yy} + u_{zz}.$$

The operator Δ is called the *Laplacian*. It is one of the central operators in applied mathematics.

Interpretation of the operators

- ∇u points in the direction of fastest increase of u .
- $\nabla \cdot F$ measures local expansion or compression of a vector field.
- $\nabla \times F$ measures local rotation.
- Δu compares a value with nearby values and measures local curvature or imbalance.

Example 2.1: Gradient, divergence, curl, and Laplacian

Let $u(x, y, z) = x^2 + y^2 + z^2$. Then

$$\nabla u = \langle 2x, 2y, 2z \rangle, \quad \Delta u = u_{xx} + u_{yy} + u_{zz} = 2 + 2 + 2 = 6.$$

If $F(x, y, z) = \langle x, y, z \rangle$, then

$$\nabla \cdot F = 1 + 1 + 1 = 3, \quad \nabla \times F = \langle 0, 0, 0 \rangle.$$

This vector field expands outward from the origin but does not rotate.

Contours and gradient field of $u = x^2 + 0.5y^2$

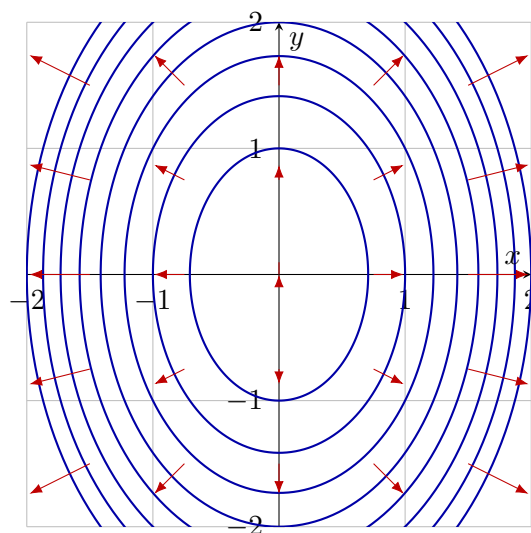


Figure 2: A scalar field, its contours, and its gradient vectors. The gradient points uphill and is perpendicular to level curves.

3 Balance laws and conservation

Many applied models begin with a balance principle:

$$\text{rate of change inside a region} = \text{inflow} - \text{outflow} + \text{sources}.$$

Suppose $\rho(x, y, z, t)$ is the density of a quantity and $J(x, y, z, t)$ is its flux field. For a fixed region E , the total amount inside E is

$$\int_E \rho dV.$$

The outward flux through the boundary ∂E is

$$\iint_{\partial E} J \cdot n dS.$$

If there are no internal sources, conservation says

$$\frac{d}{dt} \int_E \rho dV = - \iint_{\partial E} J \cdot n dS.$$

Using the Divergence Theorem,

$$\iint_{\partial E} J \cdot n dS = \int_E \nabla \cdot J dV.$$

Therefore

$$\int_E \rho_t dV = - \int_E \nabla \cdot J dV.$$

Since this should hold for every region E , the local form is

$$\rho_t + \nabla \cdot J = 0.$$

This is called a *continuity equation*.

Conservation law template

A large class of applied models has the form

$$\rho_t + \nabla \cdot J = s,$$

where ρ is a density, J is a flux, and s is a source term.

Example 3.1: Source, sink, and incompressible fields

In the plane, consider

$$F_1(x, y) = \langle x, y \rangle, \quad F_2(x, y) = \langle -x, -y \rangle, \quad F_3(x, y) = \langle -y, x \rangle.$$

Their divergences are

$$\nabla \cdot F_1 = 2, \quad \nabla \cdot F_2 = -2, \quad \nabla \cdot F_3 = 0.$$

The first field is a source, the second is a sink, and the third is rotation without local expansion.

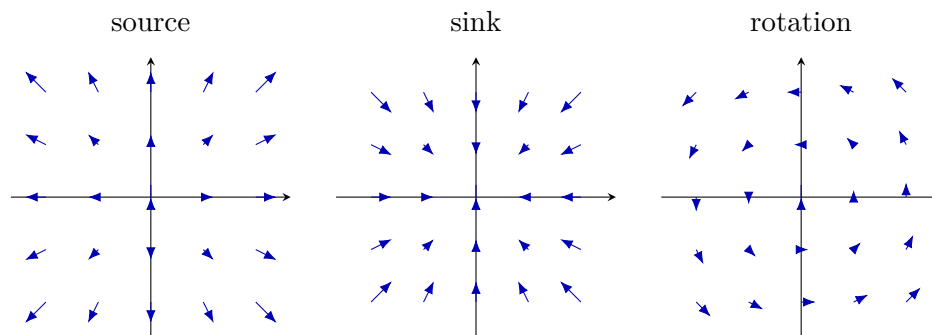


Figure 3: Three basic vector fields: source, sink, and rotational field. Divergence distinguishes expansion from compression.

4 Diffusion and the heat equation

One of the most important applied models is the heat equation. Suppose $u(x, t)$ is temperature along a rod. Heat flows from high temperature to low temperature. A common model for heat flux is Fourier's law:

$$J = -k\nabla u,$$

where $k > 0$ is the conductivity. The minus sign means heat flows downhill, opposite the direction of increasing temperature.

If heat is conserved and $J = -k\nabla u$, then

$$u_t + \nabla \cdot J = 0.$$

Substituting the flux law gives

$$u_t - k\nabla \cdot (\nabla u) = 0,$$

so

$$u_t = k\Delta u.$$

This is the *heat equation*. In one spatial dimension, $u_t = ku_{xx}$. In two spatial dimensions, $u_t = k(u_{xx} + u_{yy})$.

Why the Laplacian appears

The Laplacian measures whether a point is higher or lower than its neighbors. If a point is hotter than its neighbors, heat diffuses away. If it is colder than its neighbors, heat diffuses in. This local averaging behavior is exactly what the heat equation models.

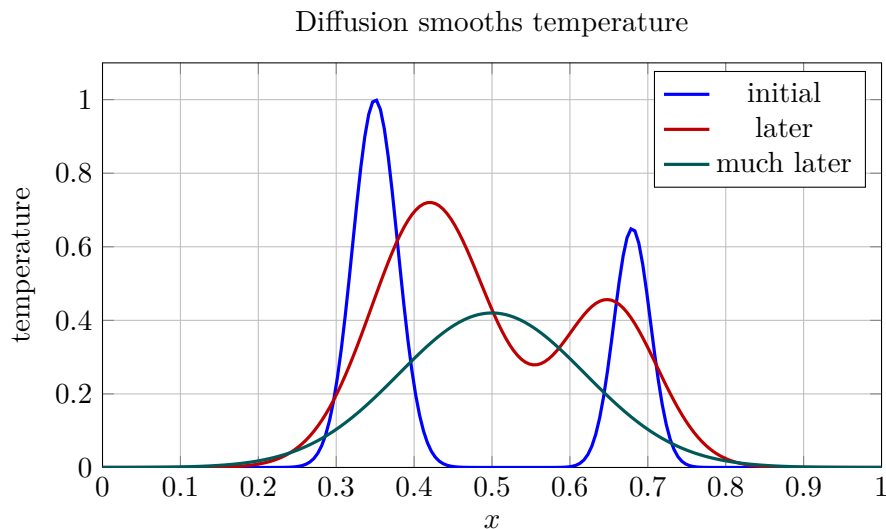


Figure 4: A localized temperature profile spreads and smooths over time, as predicted by the one-dimensional heat equation.

5 The Laplacian as local averaging

On a grid, the two-dimensional Laplacian is often approximated by

$$\Delta u(x_i, y_j) \approx \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}}{h^2}.$$

This formula compares a grid value to its four nearest neighbors. It appears in numerical partial differential equations, image processing, graph learning, and spatial statistics.

	$u_{i,j+1}$	
$u_{i-1,j}$	$u_{i,j}$	$u_{i+1,j}$
	$u_{i,j-1}$	

$$\Delta u \approx \frac{\text{neighbor sum} - 4 \cdot \text{center}}{h^2}$$

Figure 5: The five-point stencil for the two-dimensional discrete Laplacian.

6 Advection and transport

Diffusion describes spreading. Advection describes transport by a velocity field. Suppose $\rho(x, y, t)$ is the density of a substance moving with velocity field

$$v(x, y) = \langle P(x, y), Q(x, y) \rangle.$$

The flux is $J = \rho v$, and the conservation law becomes

$$\rho_t + \nabla \cdot (\rho v) = 0.$$

If the velocity field is incompressible, meaning $\nabla \cdot v = 0$, then

$$\rho_t + v \cdot \nabla \rho = 0.$$

The term $v \cdot \nabla \rho$ is a directional derivative: it measures how density changes in the direction of the flow.

Example 6.1: Rotation transports density

For $v(x, y) = \langle -y, x \rangle$, particles rotate around the origin. Since

$$\nabla \cdot v = \frac{\partial(-y)}{\partial x} + \frac{\partial x}{\partial y} = 0,$$

this field preserves area locally.

Advection by $v = \langle -y, x \rangle$

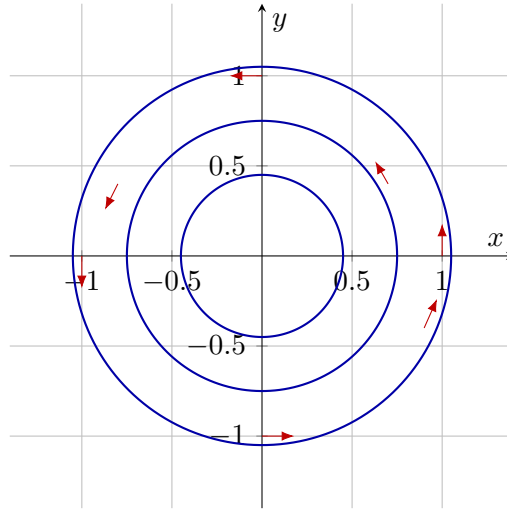


Figure 6: Particle trajectories in a rotational velocity field. Advection transports quantities along flow lines.

7 Steady states and Laplace's equation

A steady state is a solution that does not change with time. For the heat equation $u_t = k\Delta u$, a steady state satisfies

$$\Delta u = 0.$$

This is *Laplace's equation*. Functions satisfying $\Delta u = 0$ are called *harmonic functions*. Laplace's equation appears in heat conduction, electrostatics, gravitational potential, fluid flow, and image interpolation.

Example 7.1: A harmonic function

Let $u(x, y) = x^2 - y^2$. Then $u_{xx} = 2$ and $u_{yy} = -2$, so

$$\Delta u = u_{xx} + u_{yy} = 0.$$

Thus $u(x, y) = x^2 - y^2$ is harmonic.

Level curves and gradient field for $u = x^2 - y^2$

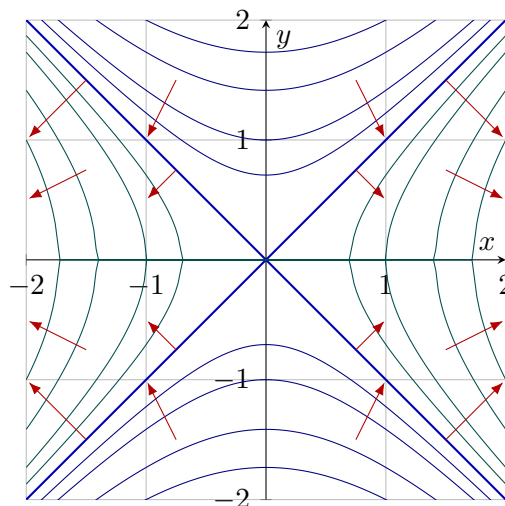


Figure 7: The harmonic function $u = x^2 - y^2$ has zero Laplacian. It bends upward in one direction and downward in the other.

8 Numerical derivatives on grids

In applied mathematics, functions are often known only through sampled values. For grid spacing h , common finite-difference approximations are

$$u_x(x_i, y_j) \approx \frac{u_{i+1,j} - u_{i-1,j}}{2h}, \quad u_y(x_i, y_j) \approx \frac{u_{i,j+1} - u_{i,j-1}}{2h},$$

$$\Delta u(x_i, y_j) \approx \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}}{h^2}.$$

These approximations convert calculus into linear algebra. A differential operator becomes a matrix or a stencil.

Continuous-discrete dictionary

Continuous calculus	Discrete computation
function $u(x, y)$	array of values
derivative u_x	finite difference
gradient ∇u	pair of arrays
Laplacian Δu	stencil or sparse matrix
integral	sum times area element
partial differential equation	update rule or linear system

9 Variational principles and energy minimization

Many equations in applied mathematics come from minimizing an energy. A simple example is the Dirichlet energy

$$E[u] = \frac{1}{2} \int_D |\nabla u|^2 dA.$$

A function that minimizes this energy subject to fixed boundary values satisfies Laplace's equation

$$\Delta u = 0$$

inside the domain.

The intuition is that minimizing $\int |\nabla u|^2$ favors smooth functions. The minimizer is as smooth as possible while still matching boundary constraints. This idea appears in elastic membranes, electrostatic potentials, image denoising and interpolation, smoothing splines, graph-based semi-supervised learning, and regularization in inverse problems.

Example 9.1: Smoothing by energy minimization

Suppose we observe noisy data y_i and want a smooth curve u_i . A discrete energy is

$$E(u) = \sum_i (u_i - y_i)^2 + \lambda \sum_i (u_{i+1} - u_i)^2.$$

The first term keeps u close to the data. The second term penalizes roughness. The parameter λ controls the tradeoff.

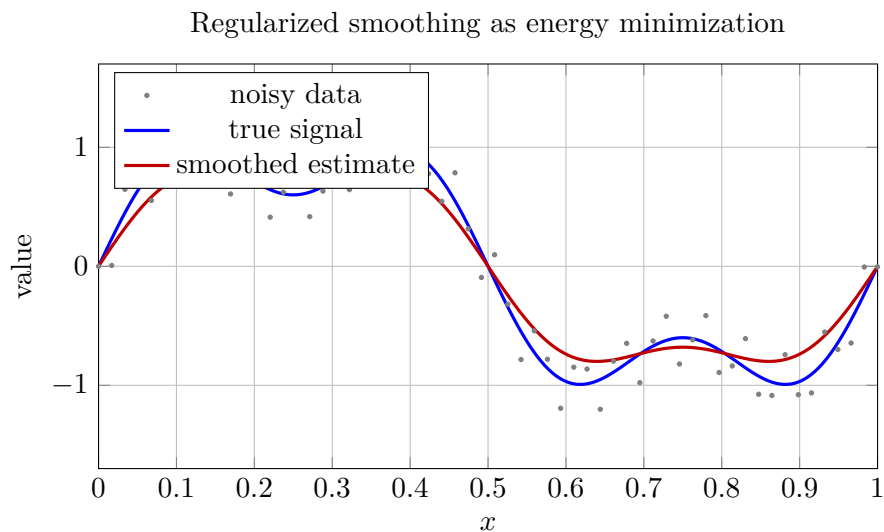


Figure 8: Energy-based smoothing balances fidelity to noisy data with a penalty on roughness.

10 Gradient flows

Optimization can also be viewed as an applied dynamical system. Given an energy or loss function $E(x)$, gradient flow is the differential equation

$$\frac{dx}{dt} = -\nabla E(x).$$

This equation moves downhill as quickly as possible in continuous time. Gradient descent is a discrete approximation:

$$x_{k+1} = x_k - \eta \nabla E(x_k).$$

Thus optimization algorithms are closely related to differential equations.

Example 10.1: Gradient flow toward a minimum

For an energy landscape $E(x, y)$, gradient flow follows the vector field $-\nabla E$. The motion crosses contours in the downhill direction and tends to move toward local minima.

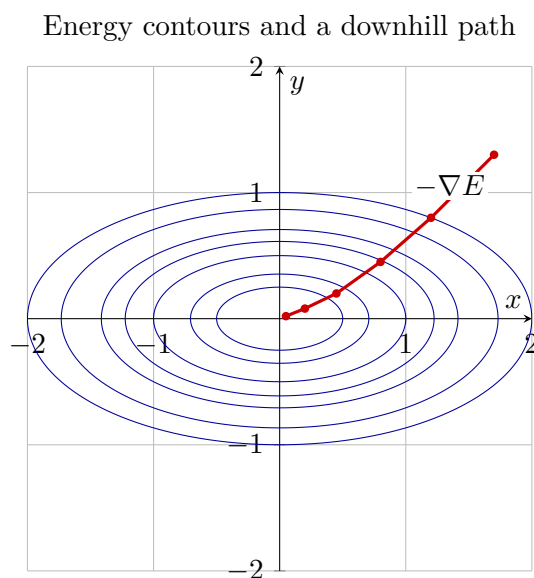


Figure 9: Gradient flow follows the vector field $-\nabla E$ and moves downhill on an energy landscape.

11 From continuous models to numerical algorithms

Applied mathematics often converts a continuous model into a finite-dimensional computation. The common steps are:

1. Choose variables. Decide what the unknown function represents.
2. State a law. Use balance, geometry, probability, or energy.
3. Write equations. Express the law using derivatives and integrals.
4. Choose boundary and initial conditions. Models need enough information to determine a solution.

5. Discretize. Replace derivatives by finite differences, finite elements, or spectral approximations.
6. Compute. Solve a system of algebraic equations or run an update rule.
7. Validate. Compare against data, units, limiting cases, or known solutions.

For the one-dimensional heat equation $u_t = ku_{xx}$, a simple explicit method is

$$u_i^{n+1} = u_i^n + r(u_{i+1}^n - 2u_i^n + u_{i-1}^n), \quad r = \frac{k\Delta t}{(\Delta x)^2}.$$

This formula says: update each value using the difference between the value and its neighbors.

Stability matters

A numerical formula can be mathematically reasonable but computationally unstable. For the explicit heat equation scheme, the time step must be sufficiently small relative to the spatial grid size. This is a first glimpse of numerical analysis.

12 Applied mathematics examples

Image denoising

An image can be viewed as a scalar field $I(x, y)$. Diffusion smooths the image by reducing sharp local variations. A simple denoising model is $I_t = \Delta I$. This smooths noise but may also blur edges. More advanced models modify the diffusion coefficient so smoothing happens more strongly in flat regions and less strongly near edges.

Probability diffusion

A probability density $p(x, t)$ can evolve by diffusion:

$$p_t = Dp_{xx}.$$

Under appropriate boundary conditions this equation preserves total probability:

$$\int p(x, t) dx = 1.$$

This connects calculus to stochastic processes, Brownian motion, and statistical physics.

Inverse problems

In an inverse problem, we infer hidden causes from observed effects. A common regularized model minimizes

$$\|Au - y\|^2 + \lambda\|Lu\|^2,$$

where A models the measurement process, y is observed data, and L measures roughness.

Conservation in networks

Even when a system is not spatially continuous, conservation ideas remain. In a network, mass or probability may flow along edges. The graph divergence at a node measures net outflow. The graph Laplacian plays the role of the continuous Laplacian. This is why Laplacian ideas appear in spectral clustering, graph signal processing, semi-supervised learning, and diffusion maps.

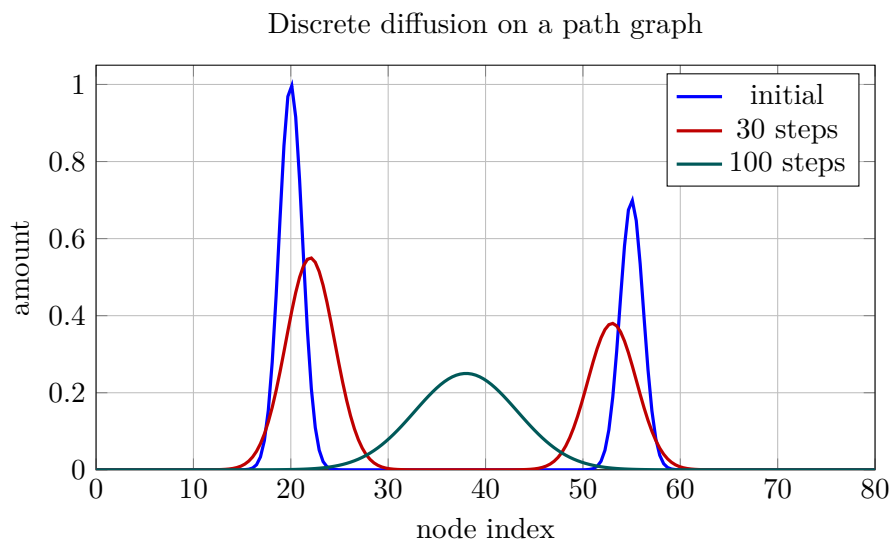


Figure 10: Discrete diffusion on a one-dimensional network. Repeated averaging behaves like a graph version of the heat equation.

13 Common modeling mistakes

Applied mathematics is powerful, but models require judgment. Common mistakes include:

- confusing density with total amount;
- forgetting units when defining flux or source terms;
- using a local equation without appropriate boundary conditions;
- assuming a numerical solution is accurate without checking stability or convergence;
- treating a model as reality rather than as a simplified representation;
- ignoring whether a vector field is conservative, divergence-free, or rotational;
- applying a theorem without checking orientation, smoothness, or region assumptions.

A good model should be mathematically consistent, computationally feasible, and interpretable in context.

14 Computational lab: finite differences and diffusion

The following functions form a small numerical calculus toolkit for scalar fields on a grid:

```
def central_gradient(U, h):
    Ux[:,1:-1] = (U[:,2:] - U[:, :-2])/(2*h)
    Uy[1:-1,:] = (U[2:,:] - U[:-2,:])/(2*h)
    return Ux, Uy

def five_point_laplacian(U, h):
    Lap[1:-1,1:-1] = (
        U[2:,1:-1] + U[:-2,1:-1]
        + U[1:-1,2:] + U[1:-1,:-2]
        - 4*U[1:-1,1:-1]
    )/(h**2)
    return Lap
```

For $u(x, y) = x^2 + y^2$, the exact value is $\Delta u = 4$. The numerical value should be close to 4 on interior grid points.

15 Chapter summary

This chapter connected multivariable calculus to applied mathematics. The central ideas are:

- scalar fields describe quantities such as temperature, density, potential, image intensity, and loss;
- vector fields describe forces, velocities, fluxes, and gradient flows;
- conservation laws combine time derivatives with flux integrals;
- the Divergence Theorem converts global flux balance into local differential equations;
- diffusion and the heat equation arise from conservation plus a flux law;
- the Laplacian measures local imbalance and acts like a continuous averaging operator;
- finite differences turn derivatives into computational formulas;
- energy minimization leads to differential equations and regularized algorithms;
- gradient flow connects optimization with dynamical systems.

The main conceptual bridge is

$$\text{calculus operators} \longrightarrow \text{models} \longrightarrow \text{algorithms}.$$

16 Exercises

Conceptual exercises

1. Explain the difference between a scalar field and a vector field. Give two examples of each from applied mathematics.
2. What does ∇u measure? Why is it perpendicular to level curves?

3. What does $\nabla \cdot F$ measure physically?
4. What does Δu measure intuitively?
5. Explain why heat should flow in the direction $-\nabla u$ rather than ∇u .
6. What is the difference between diffusion and advection?
7. Why does a conservation law naturally involve flux through the boundary of a region?
8. Explain why a steady-state heat equation leads to $\Delta u = 0$.
9. Why do numerical models require boundary conditions?
10. Explain the connection between gradient descent and gradient flow.

Computational exercises

11. For $u(x, y) = e^{-x^2-y^2}$, compute ∇u and Δu .
12. For $F(x, y) = \langle x^2, xy \rangle$, compute $\nabla \cdot F$ and interpret the result.
13. For $F(x, y) = \langle -y, x \rangle$, compute divergence and curl.
14. Use finite differences to approximate u_x and u_y for $u(x, y) = \sin x \cos y$ on a grid.
15. Use the five-point stencil to approximate Δu for $u(x, y) = x^2 + y^2$.
16. Simulate one-dimensional diffusion with a different initial condition.
17. Simulate two-dimensional diffusion from a square-shaped hot region.
18. Compare numerical diffusion using two different time step sizes.
19. Implement gradient descent for $E(x, y) = x^2 + 4y^2$ and plot the path.
20. Solve a smoothing problem using $E(u) = \sum_i (u_i - y_i)^2 + \lambda \sum_i (u_{i+1} - u_i)^2$ for several values of λ .

Applied exercises

21. Derive the heat equation from $\rho_t + \nabla \cdot J = 0$ and Fourier's law $J = -k \nabla u$.
22. Suppose $p(x, t)$ is a probability density satisfying $p_t = D p_{xx}$ on a closed interval with no-flux boundary conditions. Explain why total probability is conserved.
23. A pollutant concentration $c(x, y, t)$ moves with velocity $v(x, y)$ and diffuses with coefficient D . Explain the model $c_t + v \cdot \nabla c = D \Delta c$ term by term.
24. In image processing, why might ordinary diffusion blur edges? How might one modify the model to reduce edge blurring?
25. In a loss landscape $L(\theta)$, explain the difference between gradient flow and Newton's method.

Challenge exercises

26. Let $u(x, y) = \log \sqrt{x^2 + y^2}$ for $(x, y) \neq (0, 0)$. Show that $\Delta u = 0$ away from the origin.
27. Let $F = \nabla u$. Show that $\nabla \cdot F = \Delta u$.
28. Let $v(x, y) = \langle -y, x \rangle$. Show that the flow preserves $x^2 + y^2$.
29. Derive the finite-difference update $u_i^{n+1} = u_i^n + r(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$ from $u_t = ku_{xx}$.
30. For $E(u) = \|u - y\|^2 + \lambda \|Du\|^2$, show that the minimizer satisfies $(I + \lambda D^T D)u = y$.